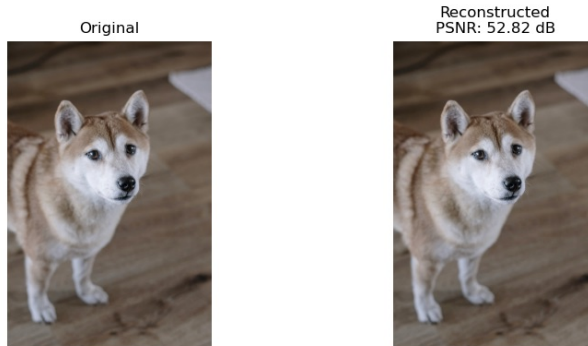


(1) Write a Matlab or Python code for the 4:2:0 image compression technique.

$B = C420(A)$, where A is the input color image and B is the reconstructed image. Just use the interpolation method for reconstruction. The code should be handed out by [NTUCool](#).

(Note: The command `rgb2ycbcr` cannot be used).

Moreover, please use the PSNR to compute the difference between A and B . (25 scores)



(2) In the JPEG process, (a) why the DCT is used instead of the DFT for transformation? Write at least two reasons. (b) Why the input image is separated into several 8×8 blocks before using the DCT? Write at least two reasons. (10 scores)

- (a)
1. DCT operates only on real number, and produces real output value, while DFT generally produces complex numbers.
 2. DCT concentrating image energy into low frequency components, which allows for better compression with less perceptible loss.
- (b)
1. The frequency of an image varies with the location.
 2. Saving the memory
 3. Less complexity $\rightarrow O(MN)$

- (3) (a) Give at least three conditions where the two images look similar but the NRMSE is large.
 (b) Give at least two conditions where the two vocal signals sound similar but the NRMSE is large.
 (10 scores)

(a) 1. Global brightness, place with large differences will be magnified.
 2. Slight spatial misalignment.
 3. Small rotation or warping.

(b) 1. When signal is delayed of the other, samples are not aligned.
 2. One signal is slightly louder or processed with equalization.

- ✓(4) Suppose that $P(x = n) = \frac{(1 - e^{-\lambda}) \exp(-\lambda n)}{1 - e^{-\lambda}}$ for $n = 0, 1, 2, 3, \dots, 5000$ where $\lambda = 0.015$.
 Also suppose that $\text{length}(x) = 50000$. Estimate the range of the total coding lengths in the binary system when using (i) the Huffman code and (ii) the arithmetic code.
 (15 scores)

$$\begin{aligned}
 \text{entropy} &= - \sum_{j=1}^J P(S_j) \ln P(S_j) \\
 &= - \sum_{n=1}^{5000} (1 - e^{-0.015}) e^{-0.015n} \ln (1 - e^{-0.015}) e^{-0.015n} \\
 &= - \sum_{n=1}^{5000} (1 - e^{-0.015}) e^{-0.015n} [\ln (1 - e^{-0.015}) - 0.015n] \\
 &= \underbrace{- \ln(1 - e^{-0.015}) \sum_{n=1}^{5000} (1 - e^{-0.015}) e^{-0.015n}}_1 + \underbrace{0.015 \sum_{n=1}^{5000} (1 - e^{-0.015}) e^{-0.015n} \cdot n}_{\frac{e^{-\lambda}}{1 - e^{-\lambda}}} \\
 &= -(-4.206) + 0.015 \times 66.11 \\
 &= 5.198 \quad *
 \end{aligned}$$

$$\begin{aligned}
 (1 - e^{-\lambda}) \sum_{n=1}^{\infty} e^{-\lambda n} &= 1 \\
 (1 - e^{-\lambda}) \sum_{n=1}^{\infty} e^{-\lambda n} \cdot n &= \frac{e^{-\lambda}}{1 - e^{-\lambda}}
 \end{aligned}$$

(i) huffman :

$$\begin{aligned}
 \text{ceil} \left(5 \times 10^4 \times \frac{5.198}{\ln 2} \right) &\leq \text{length} \leq \text{floor} \left(5 \times 10^4 \times \frac{5.198}{\ln 2} + 5 \times 10^4 \right) \\
 &\approx 374956 \leq \text{length} \leq 424956 \quad *
 \end{aligned}$$

(ii) arithmetic :

$$\begin{aligned}
 \text{ceil} \left(5 \times 10^4 \times \frac{5.198}{\ln 2} \right) &\leq \text{length} \leq \text{floor} \left(5 \times 10^4 \times \frac{5.198}{\ln 2} + \log_2 2 + 1 \right) \\
 &\approx 374956 \leq \text{length} \leq 374958 \quad *
 \end{aligned}$$

(5) How do we implement the following matrix operation with the least number of multiplications? (10 scores)

$$\begin{bmatrix} y_0 \\ y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0.7010 & 0.7010 & 0.7010 & 0.7010 \\ 0.9239 & 0.3827 & -0.3827 & -0.9239 \\ 0.7010 & -0.7010 & -0.7010 & 0.7010 \\ 0.3827 & -0.9239 & 0.9239 & -0.3827 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{aligned} a &= x_0 + x_3 & y_0 &= A = 0.7010(a + b) & \rightarrow & 1 \text{ MUL} \\ b &= x_1 + x_2 & y_1 &= B = 0.9239c + 0.3827d & \rightarrow & 2 \text{ MUL} \\ c &= x_0 - x_3 & y_2 &= C = 0.7010(a - b) & \rightarrow & 1 \text{ MUL} \\ d &= x_1 - x_2 & y_3 &= D = 0.9239c - 0.3827d & \rightarrow & 2 \text{ MUL} \end{aligned}$$

total 6 Multiplications. *

(6) For the rotation operation as follows:

$$\begin{bmatrix} y_0 \\ y_1 \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \end{bmatrix}$$

In what conditions we can implement it with only two multiplications? Express the solutions in terms of . (10 scores)

(i) if $\sin \theta = \pm \cos \theta$, $\theta = \frac{n\pi}{4}$, $n = 1, 3, 5, \dots$

$$\bullet \sin \theta = \cos \theta$$

$$\bullet \sin \theta = -\cos \theta$$

$$\begin{bmatrix} y_0 \\ y_1 \end{bmatrix} = \begin{bmatrix} \cos \theta & \cos \theta \\ \cos \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -2\cos \theta & 0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} \quad \begin{bmatrix} y_0 \\ y_1 \end{bmatrix} = \begin{bmatrix} \cos \theta & \cos \theta \\ \cos \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} + \begin{bmatrix} 0 & -2\cos \theta \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \end{bmatrix}$$

(ii) if $\sin \theta = \pm z^{-k}$, $\theta = \sin^{-1}(\pm z^{-k})$, $k = 1, 2, 3, \dots$

$$\begin{bmatrix} y_0 \\ y_1 \end{bmatrix} = \begin{bmatrix} \cos \theta & \pm z^{-k} \\ \pm z^{-k} & \cos \theta \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} \Rightarrow \begin{aligned} y_0 &= x_0 \cos \theta \pm x_1 z^{-k} \\ y_1 &= \pm x_0 z^{-k} + x_1 \cos \theta \end{aligned}$$

(iii) if $\cos \theta = \pm z^{-k}$, $\theta = \cos^{-1}(\pm z^{-k})$, $k = 1, 2, 3, \dots$

$$\begin{bmatrix} y_0 \\ y_1 \end{bmatrix} = \begin{bmatrix} \pm z^{-k} & \sin \theta \\ -\sin \theta & \pm z^{-k} \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} \Rightarrow \begin{aligned} y_0 &= \pm x_0 z^{-k} + x_1 \sin \theta \\ y_1 &= -x_0 \sin \theta \pm x_1 z^{-k} \end{aligned}$$

(iv) if $\sin \theta = \pm z^{\pm k} \cos \theta$, $k = 1, 2, 3, \dots$

$$\begin{bmatrix} y_0 \\ y_1 \end{bmatrix} = \begin{bmatrix} \cos \theta & \pm z^{\pm k} \cos \theta \\ \pm z^{\pm k} \cos \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_0 \\ x_1 \end{bmatrix} \Rightarrow \begin{aligned} y_0 &= \cos \theta (x_0 \pm x_1 z^{\pm k}) \\ y_1 &= \cos \theta (\pm x_0 z^{\pm k} + x_1) \end{aligned}$$

(7) Consider the implementation efficiency, what is the optimal number of points of the DFTs if we want to implement the convolution of $x[n]$ and $h[n]$ when $\text{length}(x[n]) = 63$ and $\text{length}(h[n]) = 35$? (5 scores)

$$P \geq M+N-1$$

$$P \geq 63+35-1 \quad 63+35-1 = 97, \text{ 取 } 96.$$

$$96 - \text{point of DFT.}$$

✓(8) Determine the number of real multiplications for (a) the 154-point DFT, (b) the 165-point DFT, and (c) the 242-point DFT. (15 scores)

(a) 154 point DFT.

$$\begin{aligned} \text{MUL}_{154} &= 11 \text{ MUL}_{14} + 14 \text{ MUL}_{11} \\ &= 11(2 \text{ MUL}_7 + 7 \text{ MUL}_2) + 14 \text{ MUL}_{11} \\ &= 22 \times 16 + 14 \times 32 \\ &= 800 \end{aligned}$$

(b) 165 point DFT

$$\begin{aligned} \text{MUL}_{165} &= 11 \text{ MUL}_{15} + 15 \text{ MUL}_{11} \\ &= 11 \times 40 + 15 \times 40 \\ &= 1040 \end{aligned}$$

(c) 242 point DFT.

$$\begin{aligned} \text{MUL}_{242} &= 2 \text{ MUL}_{121} + 121 \text{ MUL}_2 \\ &= 2(11 \text{ MUL}_{11} + 11 \text{ MUL}_{11} + 3 D_1) \\ &= 2 \times (2 \times 11 \times 40 + 3 \times 10 \times 10) \\ &= 2360 \end{aligned}$$

Extra: $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} a & a \\ a & b \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ MUL ? ADD ?

$$= \begin{bmatrix} a & a \\ a & a \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & b-a \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} a & a \\ a & a \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \begin{bmatrix} z_3 \\ z_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & b-a \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

• $z_1 = z_2 = a(x_1 + x_2)$ • $z_4 = (b-a)x_2$

$$y_1 = z_1$$

• $y_2 = z_2 + z_4$

⇒ 2 MULs, 2 ADDs. *